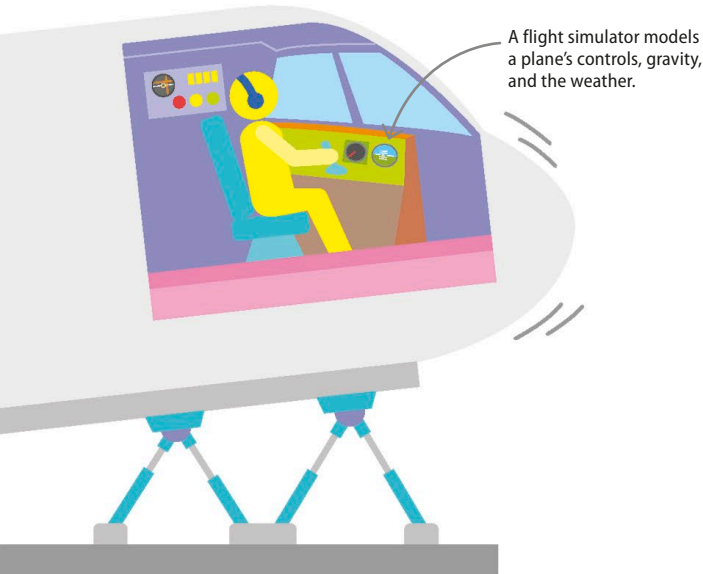


Making a model

A model is a representation of a real-world system or object. It has the object's main features, and is easy to recognize, but it's clearly not the real thing. For instance, a model of the Eiffel Tower might be really small and made of blue plastic, but is still recognizable as the Eiffel Tower. In a similar way, a computer model of an object or system won't have a lot of the details, but it should be a recognizable representation of it – it is an abstraction of the real thing.

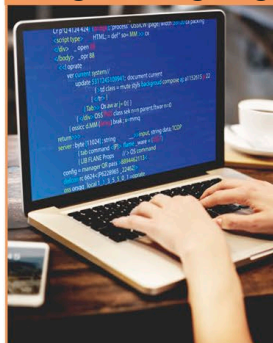


△ Why make a model?

A real-world model allows users to analyse the system without damaging it or endangering themselves. A flight simulator allows pilots to practise handling a variety of difficult and dangerous situations without any real danger. Likewise, a computer model allows programmers to figure out what solutions might work for the problem at hand.

TOP TECH

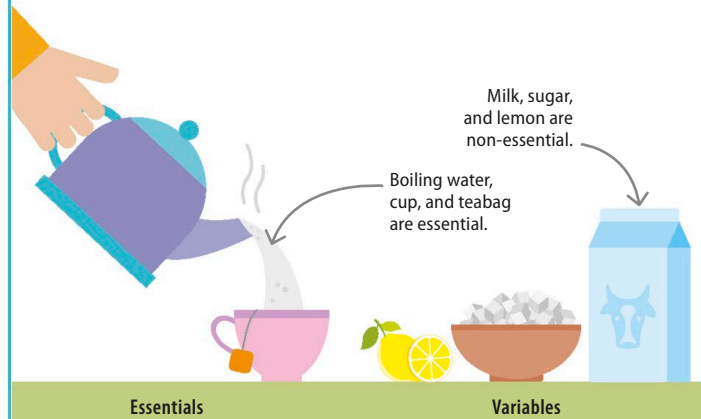
Programming languages



Earlier programming languages required programmers to move values around the computer's internal storage. Modern languages, on the other hand, are examples of abstraction. They allow programmers to instruct computers in a more natural way, without knowing exactly what's going on inside.

Identifying variables

A model based on only the essential parts of an object or system limits the amount of analysis possible. An important step is to identify other details that change how the model behaves as the value of the details change. These non-essential details are called variables because their values vary rather than staying the same. Computer scientists have to work out which non-essential details in a system actually make a difference to the model's behaviour. These can then be added to the model.



△ Tea variables

In a model cup of tea, a cup, boiling water, and a teabag or tea leaves are essential. The amounts of sugar, milk, or lemon are variables. The outcome depends on how much sugar, milk, or lemon is added.

"All art is an abstraction to some degree."

Henry Moore (1898–1986), English artist

REAL WORLD

Hidden networks



Stripping away non-essential details can show that two seemingly different problems can actually be the same. Social media networks seem new, but they are like any other network, simply points linked by paths. This means that existing programming solutions can be used for social network tasks.